

Python Practice Questions with Hints/Algorithm

1. Add two numbers

- Input two numbers.
- Use + operator.
- Print the result.

2. Subtract two numbers

- Input two numbers.
- Use - operator.
- Print the result.

3. Multiply two numbers

- Input two numbers.
- Use * operator.
- Print the result.

4. Divide two numbers

- Input two numbers.
- Use / operator.
- Print the result.

5. Remainder of two numbers

- Input two numbers.
- Use % operator.
- Print the remainder.

6. Square of a number

- Input a number.
- Multiply the number by itself or use **2.
- Print the result.

7. Cube of a number

- Input a number.
- Multiply the number by itself twice or use **3.
- Print the result.

8. Sum of three numbers

- Input three numbers.
- Add them using +.

- Print the sum.

9. Average of three numbers

- Input three numbers.
- Sum the numbers and divide by 3.
- Print the result.

10. Area of a rectangle

- Input length and breadth.
- Multiply `length × breadth`.
- Print the area.

11. Perimeter of a rectangle

- Input length and breadth.
- Formula: $2 \times (\text{length} + \text{breadth})$.
- Print the perimeter.

12. Area of a square

- Input the side.
- Formula: `side ** 2`.
- Print the area.

13. Perimeter of a square

- Input the side.
- Formula: $4 \times \text{side}$.
- Print the perimeter.

14. Area of a circle

- Input radius.
- Formula: $3.14 \times \text{radius}^2$.
- Print the area.

15. Circumference of a circle

- Input radius.
- Formula: $2 \times 3.14 \times \text{radius}$.
- Print the circumference.

16. Celsius to Fahrenheit

- Input Celsius.
- Formula: $F = (C \times 9/5) + 32$.
- Print Fahrenheit.

17. Fahrenheit to Celsius

- Input Fahrenheit.
- Formula: $C = (F - 32) \times 5/9$.
- Print Celsius.

18. Simple Interest

- Input Principal (P), Rate (R), Time (T).
- Formula: $SI = (P \times R \times T) / 100$.
- Print the interest.

19. Compound Interest

- Input Principal (P), Rate (R), Time (T).
- Formula: $CI = P \times (1 + R/100)^T - P$.
- Print the result.

20. Swap two numbers

- Input two numbers a and b.
- Use `a, b = b, a` to swap.
- Print a and b.

21. Area of a triangle

- Input base and height.
- Formula: $0.5 \times \text{base} \times \text{height}$.
- Print the area.

22. Perimeter of a triangle

- Input three sides.
- Formula: `side1 + side2 + side3`.
- Print perimeter.

23. Square root of a number

- Input a number.
- Use `number ** 0.5`.
- Print the result.

24. Remainder when divided by 5

- Input a number.
- Use `% 5` to get remainder.
- Print it.

25. Sum of digits of a 2-digit number

- Input number.
- Find tens: `num // 10`, units: `num % 10`.
- Sum them and print.

26. Sum of digits of a 3-digit number

- Input number.
- Hundreds: `num // 100`, tens: `(num // 10) % 10`, units: `num % 10`.
- Sum them and print.

27. Reverse a 2-digit number

- Input number.
- Swap tens and units: `reversed = (num % 10) * 10 + (num // 10)`.
- Print reversed number.

28. Average of five numbers

- Input 5 numbers.
- Sum and divide by 5.
- Print average.

29. Sum of numbers from 1 to n

- Input n.
- Use formula: `sum = n * (n + 1) // 2`.
- Print sum.

30. Factorial of a number

- Input n.
- Initialize `fact = 1`.
- Multiply `fact` by numbers from 1 to n in a loop.
- Print factorial.

31. Area of parallelogram

- Input base and height.
- Formula: `base * height`.
- Print area.

32. Perimeter of parallelogram

- Input two sides a and b.
- Formula: `2 * (a + b)`.
- Print perimeter.

33. Area of trapezium

- Input a, b (parallel sides) and height h.
- Formula: $0.5 \times (a + b) \times h$.
- Print area.

34. Perimeter of trapezium

- Input four sides.
- Sum them.
- Print perimeter.

35. Area of rhombus

- Input diagonals d1 and d2.
- Formula: $0.5 \times d1 \times d2$.
- Print area.

36. Perimeter of rhombus

- Input side length.
- Formula: $4 \times \text{side}$.
- Print perimeter.

37. Double a number

- Input number.
- Multiply by 2.
- Print result.

38. Triple a number

- Input number.
- Multiply by 3.
- Print result.

39. Quotient and remainder of division

- Input two numbers.
- Use `//` for quotient, `%` for remainder.
- Print both.

40. Sum of squares of two numbers

- Input two numbers.
- Compute $a**2 + b**2$.
- Print sum.

41. Sum of cubes of two numbers

- Input two numbers.

- o Compute $a^{**3} + b^{**3}$.
- o Print sum.

42. Evaluate $(a + b) * c$

- o Input a, b, c.
- o Compute $(a + b) * c$.
- o Print result.

43. Evaluate $(a - b) / c$

- o Input a, b, c.
- o Compute $(a - b) / c$.
- o Print result.

44. Convert minutes into seconds

- o Input minutes.
- o Multiply by 60.
- o Print seconds.

45. Convert hours into minutes

- o Input hours.
- o Multiply by 60.
- o Print minutes.

46. Convert days into hours

- o Input days.
- o Multiply by 24.
- o Print hours.

47. Convert weeks into days

- o Input weeks.
- o Multiply by 7.
- o Print days.

48. Perimeter of a regular hexagon

- o Input side.
- o Formula: $6 \times \text{side}$.
- o Print perimeter.

49. Area of a regular hexagon

- o Input side.
- o Formula: $(3 * 1.732 / 2) * \text{side} ** 2$ ($\sqrt{3} \approx 1.732$).

- Print area.

50. **Total price with tax**

- Input item price and tax %.
 - Formula: $\text{total} = \text{price} + (\text{price} * \text{tax}/100)$.
 - Print total.
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